

Unsupervised-Domain-Adaptation Enhanced Learned Subband-based Perturbation Nonlinear Compensation for Optical Transmission Systems

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Abstract We propose a learned subband-based perturbation-aided nonlinear compensation method, enhanced by unsupervised domain adaptation using polarization- and phase-agnostic power moments. Experiments of PS-64QAM signals over 1200 km demonstrate that this method is robust to parameter mismatch and can adapt to channel variations quickly and blindly. ©2026 The Author(s)

Introduction

Fiber nonlinearity is the primary factor limiting the capacity of optical transmission systems. Traditional digital backpropagation (DBP) method suffers from high complexity [1]. Perturbation-aided nonlinear compensation (P-NLC) approaches reduce the complexity but exhibit limited performance [2]. Neural network (NN) based methods [3-4] such as learned DBP can optimize the parameters of NLC algorithms without precise link parameters. However, NN methods based on traditional NLC structures still face the challenge of balancing the performance-complexity tradeoff.

Subband-based technique offers a flexible DSP framework at the receiver [5]. It divides the received signal into multiple narrow-bandwidth subbands, performs impairments compensation, and then synthesizes the subbands. This technique has been combined with DBP [6-7], enhanced split-step Fourier method (ESSFM) [8], and P-NLC [9-10]. Due to narrow bandwidth of subbands, these methods effectively decouple chromatic dispersion (CD) and nonlinear effect, thereby reducing the nonlinear memory length and enhancing the NLC accuracy. In [8-10], single-step coupled-band ESSFM (CB-ESSFM) and subband-based P-NLC (SP-NLC) were shown to achieve significantly lower complexity and better performance than traditional NLC methods. However, these works rely on analytical equations and accurate link parameters to calculate the NLC coefficients, which is time-consuming.

In this paper, we propose a learned SP-NLC (LSP-NLC) method, enhanced by unsupervised domain adaptation (UDA) using polarization- and phase-agnostic statistical descriptors. This method avoids complex calculations and manual optimization of NLC parameters. The proposed UDA employs moments of signal power to track channel variations without a training sequence. Experiments of a wavelength-division-multiplexed probabilistically-shaped (PS) 64QAM system demonstrate that the proposed method eliminates the

need for precise link parameters and can also track channel variations quickly and blindly.

Principle of the proposed LSP-NLC with UDA

Fig. 1 shows the principle of the proposed LSP-NLC, which is designed by unrolling the steps of the SP-NLC model into a deep-learning-based computational graph. The trainable parameters θ consist of dispersion coefficient β_2 , nonlinear parameter γ_a , and perturbation coefficients $\mathbf{C}$. The forward signal flow (indicated by black arrows) begins with a fullband dual-polarization (DP) signal $\mathbf{r}_{in}(n)=[r_{x,in}(n), r_{y,in}(n)]$, where n is the index of samples. $\mathbf{r}_{in}(n)$ is decomposed into M parallel subbands in the frequency domain by a learnable linear layer. The filter $H_m(\omega|\beta_2)$ includes the frequency responses of the decomposition filter and half CDC for the m^{th} subband. Subsequently, the nonlinear phase of the subband signals, $\mathbf{u}_m(n)=[u_{x,m}(n), u_{y,m}(n)]$, is compensated [9], which is implemented by a convolutional layer with the convolution kernel $\mathbf{C}=[C_{0 \rightarrow 0}(p), \dots, C_{k \rightarrow m}(p), \dots, C_{M-1 \rightarrow M-1}(p)]$ and an active function of $f(\cdot|\gamma_a)=\exp(-j8/9\gamma_a \times (\cdot))$. After another linear layer, where $G_m(\omega|\beta_2)$ includes the frequency responses of the synthesis filter and the second half CDC, we obtain the output of the LSP-NLC $\mathbf{r}_{out}=[r_{x,out}(n), r_{y,out}(n)]$, which is then processed by butterfly filter (BF) and carrier phase recovery (CPR).

At the initial training phase, the mean squared error (MSE) between the signal after CPR and the transmitted labels $\mathbf{L}$ is calculated as the loss function. The gradient flow (indicated by blue arrows) is propagated backward through the unrolled layers using the back-propagation algorithm. The Adam optimizer is employed to update the parameters θ . Different from conventional SP-NLC or CB-ESSFM, where $\mathbf{C}$ is calculated using parameter-dependent analytical equations while γ_a and β_2 are optimized manually, this strategy enables the LSP-NLC to adaptively find the optimal parameters that minimize the distortion.

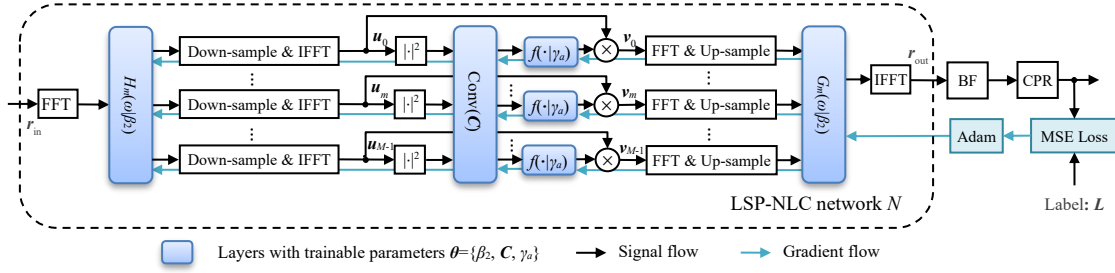


Fig. 1: The principle of the proposed LSP-NLC. BF: butterfly filter; CPR: carrier phase recovery. Label L is only used at the initial training stage and is not required during channel tracking.

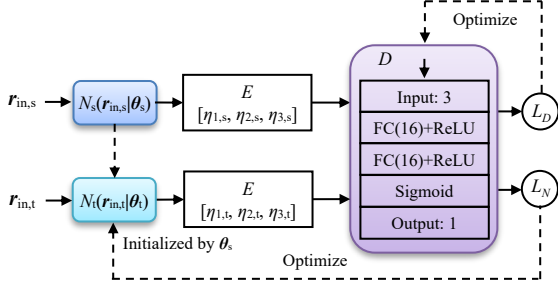


Fig. 2: The architecture of the proposed UDA. N_s and N_t are the initially-trained and tracking LSP-NLC network. E outputs polarization- and phase-agnostic statistical descriptors η .

Furthermore, a UDA method is proposed to enhance the adaptation capability of the system. We denote N_s and N_t as the initially-trained and dynamically-tracking LSP-NLC network with parameters θ_s and θ_t , respectively. $r_{in,s}$ and $r_{in,t}$ are the initial training signal and the input signal during operation. The architecture of UDA is shown in Fig. 2. It is realized via generative adversarial training, where N_t is trained to mimic N_s outputs to deceive a discriminator D .

The challenge is that $N_s(r_{in,s}|\theta_s)$ and $N_t(r_{in,t}|\theta_t)$ have different output waveforms. In addition, as the LSP-NLC network is placed before BF and CPR, $N_s(r_{in,s}|\theta_s)$ and $N_t(r_{in,t}|\theta_t)$ also exhibit different states of polarization (SOP) and phase noise. To address the challenge, we propose a power-moment feature extractor (PMFE) E to capture power-related features of the output of N_s or N_t . The PMFE extracts polarization- and phase-agnostic statistical descriptors $\eta = [\eta_1, \eta_2, \eta_3]$:

$$\eta_1 = \langle |r_{x,out}(n)|^2 + |r_{y,out}(n)|^2 \rangle \quad (1)$$

$$\eta_2 = \frac{\langle (|r_{x,out}(n)|^2 + |r_{y,out}(n)|^2)^2 \rangle}{\langle |r_{x,out}(n)|^2 + |r_{y,out}(n)|^2 \rangle^2} \quad (2)$$

$$\eta_3 = \frac{\langle (|r_{x,out}(n)|^2 + |r_{y,out}(n)|^2)^3 \rangle}{\langle |r_{x,out}(n)|^2 + |r_{y,out}(n)|^2 \rangle^3} \quad (3)$$

where $\langle \cdot \rangle$ denotes the averaging. The rationale for η is that if N_t successfully compensates CD and nonlinear impairments, the moments of its output should be the same as those of $N_s(r_{in,s}|\theta_s)$. η is immune to SOP as the total power across two polarizations remains constant in the

presence of SOP. In addition, extracting the power-related feature can also isolate the effect of phase noise.

The extracted 3-dimensional vectors η are then fed into the discriminator D , which consists of an input layer with 3 units, two hidden fully connected (FC) layers with 16 units and rectified linear unit (ReLU) activation, and an output layer with 1 unit activated by sigmoid. The discriminator outputs 1 for $E(N_s(r_{in,s}))$ and 0 for $E(N_t(r_{in,t}))$, and is updated via Eq.(4), while N_t is updated to confuse D via Eq.(5):

$$\min_D L_D = \min_D \{ \langle (D(E(N_s(r_{in,s}))) - 1)^2 \rangle + \langle D(E(N_t(r_{in,t})))^2 \rangle \} \quad (4)$$

$$\min_{N_t} L_N = \min_{N_t} \{ \langle (D(E(N_t(r_{in,t}))) - 1)^2 \rangle \} \quad (5)$$

Through adversarial training, L_D and L_N forces η of N_t align with that of N_s .

Experimental setup and results

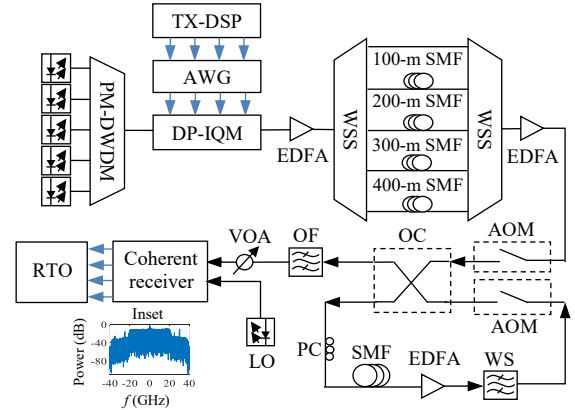


Fig. 3: Experimental setup. Inset: received spectrum.

Fig. 3 shows the experimental setup. At the transmitter, five continuous wave (CW) lights with 100-GHz channel spacing and 100-kHz laser linewidths were employed as optical carriers. The five lights were combined by a polarization-maintaining dense wavelength division multiplexer (PM-DWDM) and modulated by a 40-GBaud DP PS-64QAM signal via a 40-GHz DP I/Q modulator (DP-IQM). The PS-64QAM signal, generated from an arbitrary waveform generator (AWG), had an entropy of 5.43 bits/symbol/polarization. The modulated optical signals were amplified, separated

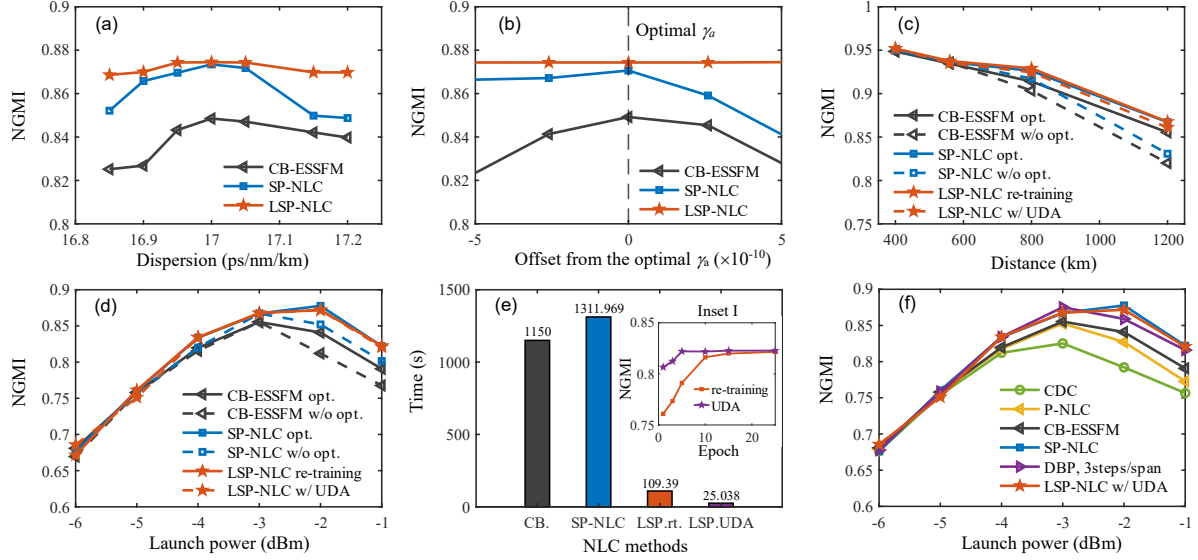


Fig. 4: (a) NGMI for different NLC methods under ambiguous CD value. (b) NGMI versus the offset from the optimal γ_a . γ_a is not the fiber nonlinear coefficient but includes this effect [9]. (c) NGMI versus distances. For dashed lines, the parameters obtained at 400 km are applied to all distances. For solid lines, the parameters are re-optimized or re-trained using a training sequence at each point. (d) NGMI versus launch power per channel. For dashed lines, the parameters obtained at -3 dBm are applied to all powers. (e) Runtime for different NLC methods. CB.: CB-ESSFM, LSP.rt.: LSP-NLC with re-training, LSP.UDA: LSP-NLC with UDA. Inset I: convergence curves of LSP-NLC using re-training and using UDA for the point of -1 dBm in Fig. 4(d). (f) NGMI versus launch power per channel. Unless otherwise stated, all figures are based on the optimal launch power and 1200 km.

by a wavelength selective switch (WSS), and decorrelated by five single-mode fiber (SMF) spans. The decorrelated signals were recombined by a WSS and launched into a recirculating loop that was implemented by an optical coupler (OC), two acoustic-optic modulators (AOMs), an 80-km SMF, an Erbium-doped fiber amplifier (EDFA), and a waveshaper (WS).

At the receiver, the central channel was selected by an optical filter (OF) and controlled by a variable optical attenuator (VOA) to set the power as -10 dBm. The optical signal was detected by a 40-GHz DP coherent receiver and sampled by a real-time oscilloscope (RTO). The Rx-DSP consisted of frequency offset compensation (CFO), matched filtering (MF), NLC, BF, and CPR. Then the normalized generalized mutual information (NGMI) was calculated.

Fig. 4(a) shows the NGMI of different NLC methods under CD ambiguity. Fig. 4(b) shows the NGMI versus offset from the optimal γ_a . As observed, the performance of CB-ESSFM and SP-NLC highly depends on the accuracy of these parameters. In contrast, LSP-NLC exhibits remarkable robustness across the entire evaluation range. This can be attributed to the “learned” nature of the architecture. When the channel parameters are ambiguous, LSP-NLC can automatically optimize the parameters.

We next assess the performance of UDA-based parameter transfer when channel conditions change. Fig. 4(c) and 4(d) compare different NLC methods under different distances and launch powers, respectively. For CB-ESSFM and SP-NLC, when the channel condition shifts away

from the initial status, as shown by the “w/o opt.” dashed lines, the performance degrades significantly compared to that using re-optimized parameters for each point (the “opt.” solid lines). However, the re-calculation of $\mathbf{C}$ is time-consuming, as shown in Fig. 4(e). On the other hand, LSP-NLC avoids analytical calculation of $\mathbf{C}$ and manual optimization of β_2 and γ_a . However, without UDA, it still needs to train the parameters using a training sequence. In contrast, LSP-NLC with UDA can consistently track channel variations across distances and powers without a training sequence and with less time.

Finally, Fig. 4(f) compares the performance of different NLC algorithms at 1200 km. Except for UDA-based LSP-NLC where the parameters were trained at -3 dBm and then transferred to other power levels, all other methods employed per-point optimization. As observed, LSP-NLC with UDA achieves performance comparable to that of 3 steps/span DBP and SP-NLC, and outperforms CB-ESSFM and P-NLC.

Conclusions

We have proposed a novel UDA-enhanced LSP-NLC method that adaptively learns NLC parameters without precise channel information. By leveraging polarization- and phase-agnostic power-moment features for adversarial training, it can quickly adapt to channel variations blindly. Experimental results demonstrate its robustness against parameter mismatch and superior tradeoff between NGMI and the adaptation time.

Acknowledgements

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